INCH-POUND
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DETAIL SPECIFICATION SHEET

RINGS, RETAINING, SPIRAL (UNIFORM CROSS SECTION) MEDIUM DUTY EXTERNAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-27426.

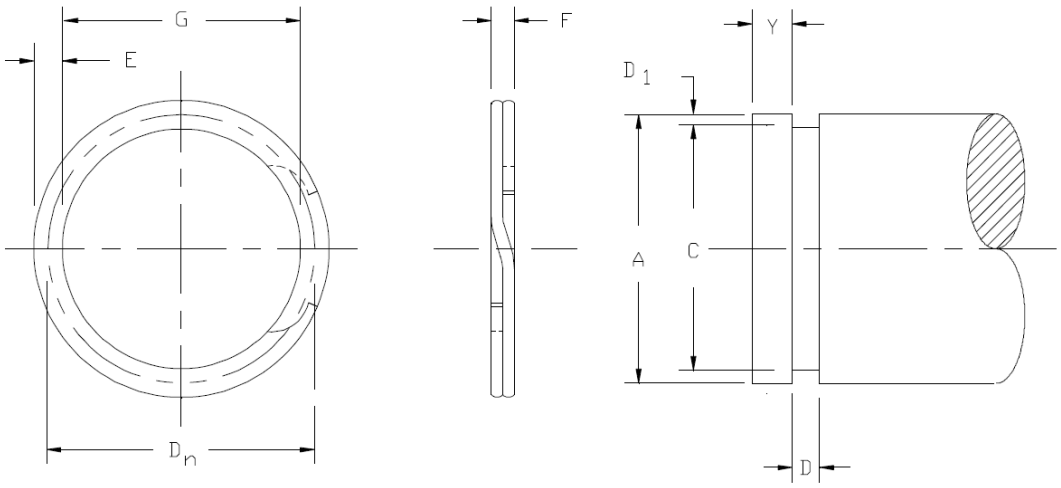


FIGURE 1. External ring.

Maximum bottom groove radius is 0.005 inches for rings up to 0.946 inches free diameter and 0.010 inches for larger rings. Ring thickness applies only to unplated rings. Add 0.002 inches upper thickness tolerance for plated rings. Groove widths listed will accept rings without modification.

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AMSC N/A FSC 5325

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TABLE I. Medium duty external retaining rings.

DASH NUMBER	SHAFT DIAMETER		RING DIMENSIONS						GROOVE DIMENSIONS				
	INCHES	MM	FREE DIAMETER		THICKNESS		RADIAL WALL		DIAMETER		WIDTH		NOMINAL GROOVE DEPTH
	A		G	Tol*	F	Tol	E	Tol	C	Tol	D	Tol	D_1
0100	0.500	12.7	0.467		0.025		0.045		0.474		0.030		0.013
0101	0.531	13.5	0.498		0.025		0.045		0.505	±0.002	0.030		0.013
0102	0.551	14.0	0.518		0.025		0.045		0.525		0.030		0.013
0103	0.562	14.3	0.529		0.025		0.045		0.536		0.030		0.013
0104	0.594	15.1	0.561		0.025		0.045		0.569		0.030		0.013
0105	0.625	15.9	0.585		0.025		0.055		0.594		0.030		0.016
0106	0.656	16.7	0.617		0.025		0.055		0.625		0.030		0.016
0107	0.669	17.0	0.629		0.025		0.055		0.638		0.030		0.016
0108	0.687	17.5	0.647		0.025		0.055		0.656		0.030		0.016
0109	0.718	18.3	0.679		0.025		0.055		0.687		0.030		0.016
0110	0.750	19.0	0.710	+0.000	0.031		0.065		0.719		0.036		0.016
0111	0.781	19.8	0.741	-0.013	0.031		0.065		0.750		0.036		0.016
0112	0.812	20.6	0.771		0.031		0.065		0.781	±0.003	0.036	+0.003	0.016
0113	0.843	21.4	0.803		0.031		0.065		0.812		0.036	-0.000	0.016
0114	0.875	22.2	0.828		0.031		0.065		0.838		0.036		0.019
0115	0.906	23.0	0.860		0.031		0.065		0.869		0.036		0.019
0116	0.937	23.8	0.889		0.031		0.065		0.900		0.036		0.019
0117	0.968	24.4	0.916		0.037		0.075		0.925		0.042		0.021
0118	0.984	25.0	0.930		0.037	±0.002	0.075		0.941		0.042		0.021
0119	1.000	25.4	0.946		0.037		0.075		0.957		0.042		0.021
0120	1.023	26.0	0.968		0.037		0.075		0.980		0.042		0.021
0121	1.031	26.3	0.978		0.037		0.075		0.988		0.042		0.021
0122	1.062	27.0	1.007		0.037		0.075	+0.003	1.020		0.042		0.021
0123	1.093	27.8	1.040		0.037		0.075	-0.005	1.051		0.042		0.021
0124	1.125	29.0	1.070		0.037		0.075		1.083		0.042		0.021
0125	1.156	29.3	1.102		0.037		0.075		1.114		0.042		0.021
0126	1.188	30.2	1.127		0.043		0.085		1.140		0.048		0.024
0127	1.218	31.0	1.159		0.043		0.085		1.170		0.048		0.024
0128	1.250	31.7	1.188		0.043		0.085		1.202		0.048		0.024
0129	1.281	32.6	1.221	+0.000	0.043		0.085		1.233	±0.004	0.048		0.024
0130	1.312	33.3	1.251	-0.015	0.043		0.095		1.264		0.048		0.024
0131	1.343	34.1	1.282		0.043		0.095		1.295		0.048		0.024
0132	1.375	34.9	1.308		0.043		0.095		1.323		0.048		0.026
0133	1.406	35.8	1.340		0.043		0.095		1.354		0.048		0.026
0134	1.437	36.5	1.370		0.043		0.095		1.385		0.048		0.026
0135	1.468	37.3	1.402		0.043		0.095		1.416		0.048		0.026
0136	1.500	38.1	1.433		0.043		0.095		1.448		0.048		0.026
0137	1.562	39.6	1.490		0.049		0.108		1.507		0.056	+0.004	0.028
0138	1.575	40.0	1.503		0.049		0.108		1.520		0.056	-0.000	0.028
0139	1.625	41.7	1.549		0.049		0.108		1.566		0.056		0.030
0140	1.687	42.8	1.610		0.049		0.118		1.628		0.056		0.030
0141	1.750	44.4	1.673		0.049		0.118		1.691		0.056		0.030
0142	1.771	44.9	1.690		0.049		0.118		1.708	±0.005	0.056		0.032
0143	1.813	46.0	1.730	+0.000	0.049		0.118		1.749		0.056		0.032
0144	1.875	47.6	1.789	-0.020	0.049		0.128		1.808		0.056		0.034
0145	1.938	49.2	1.844		0.049	±0.003	0.128		1.861		0.056		0.034
0146	1.969	50.0	1.882		0.049		0.128		1.902		0.056		0.034
0147	2.000	50.8	1.909		0.049		0.128		1.929		0.056		0.035
0148	2.062	52.3	1.971		0.049		0.128		1.992		0.056		0.035
0149	2.125	53.9	2.029		0.049		0.128		2.051		0.056		0.037
0150	2.156	54.7	2.060		0.049		0.138		2.082		0.056		0.037
0151	2.165	55.0	2.070	+0.000	0.049		0.138		2.091	±0.006	0.056		0.037
0152	2.188	55.5	2.092	-0.025	0.049		0.138		2.113		0.056		0.037
0153	2.250	57.1	2.153		0.049		0.138		2.176		0.056		0.037
0154	2.312	58.7	2.211		0.049		0.138		2.234		0.056		0.039

* = Tolerance

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TABLE I. Medium duty external retaining rings - Continued.

DASH NUMBER	SHAFT DIAMETER		RING DIMENSIONS						GROOVE DIMENSIONS				
	INCHES	MM	FREE DIAMETER		THICKNESS		RADIAL WALL		DIAMETER		WIDTH		NOMINAL GROOVE DEPTH
	A		G	Tol	F	Tol	E	Tol	C	Tol	D	Tol	D_1
0155	2.362	59.9	2.261		0.049		0.138		2.284		0.056		0.039
0156	2.375	60.3	2.273		0.049		0.138		2.297		0.056		0.039
0157	2.437	61.9	2.331	+0.000	0.049		0.148	+0.003	2.355		0.056		0.041
0158	2.500	63.5	2.394	-0.025	0.049		0.148	-0.005	2.418		0.056		0.041
0159	2.559	64.9	2.449		0.049		0.148		2.473		0.056	+0.004	0.043
0160	2.562	65.1	2.452		0.049		0.148		2.476		0.056	-0.000	0.043
0161	2.625	66.6	2.514		0.049		0.148		2.539		0.056		0.043
0162	2.688	68.2	2.572		0.049		0.158		2.597		0.056		0.045
0163	2.750	69.8	2.635		0.049		0.158		2.660		0.056		0.045
0164	2.813	71.4	2.696		0.049		0.168		2.722		0.056		0.045
0165	2.875	73.0	2.755		0.049		0.168		2.781		0.056		0.047
0166	2.937	74.5	2.817		0.049		0.168		2.843		0.056		0.047
0167	2.952	74.9	2.831	+0.000	0.049		0.168		2.858		0.056		0.047
0168	3.000	76.1	2.877	-0.030	0.061		0.168		2.904		0.068		0.048
0169	3.062	77.7	2.938		0.061		0.168		2.966		0.068		0.048
0170	3.125	79.3	3.000		0.061		0.178		3.027		0.068		0.049
0171	3.149	79.9	3.023		0.061		0.178		3.051		0.068		0.049
0172	3.187	81.0	3.061		0.061		0.178		3.089		0.068		0.049
0173	3.250	82.5	3.121		0.061		0.178		3.150		0.068		0.050
0174	3.312	84.4	3.180		0.061		0.188		3.208		0.068		0.052
0175	3.343	84.9	3.210		0.061		0.188		3.239		0.068		0.052
0176	3.375	85.8	3.242		0.061		0.188		3.271		0.068		0.052
0177	3.437	87.2	3.301		0.061	±.003	0.188		3.331		0.068		0.053
0178	3.500	88.8	3.363		0.061		0.188		3.394		0.068		0.053
0179	3.543	89.9	3.402		0.061		0.198		3.433		0.068		0.055
0180	3.562	90.5	3.422		0.061		0.198		3.452		0.068		0.055
0181	3.625	92.0	3.483		0.061		0.198		3.515		0.068		0.055
0182	3.687	93.6	3.543		0.061		0.198		3.575		0.068		0.056
0183	3.740	95.0	3.597		0.061		0.198		3.628	±0.006	0.068		0.056
0184	3.750	95.2	3.606		0.061		0.198		3.638		0.068		0.056
0185	3.812	97.1	3.668		0.061		0.198	+0.004	3.700		0.068		0.056
0186	3.875	98.3	3.724	+0.000	0.061		0.208	-0.006	3.757		0.068		0.059
0187	3.938	99.9	3.784	-0.040	0.061		0.208		3.820		0.068		0.059
0188	4.000	101.5	3.842		0.061		0.218		3.876		0.068		0.062
0189	4.063	103.2	3.906		0.061		0.218		3.939		0.068	+0.005	0.062
0190	4.125	104.8	3.967		0.061		0.218		4.000		0.068	-0.000	0.062
0191	4.134	105.0	3.975		0.061		0.218		4.010		0.068		0.062
0192	4.188	106.5	4.030		0.061		0.218		4.058		0.068		0.062
0193	4.250	107.9	4.084		0.061		0.228		4.120		0.068		0.065
0194	4.312	109.5	4.147		0.061		0.228		4.182		0.068		0.065
0195	4.331	110.0	4.164		0.061		0.228		4.200		0.068		0.065
0196	4.375	111.0	4.208		0.061		0.228		4.245		0.068		0.065
0197	4.437	112.7	4.271		0.061		0.228		4.307		0.068		0.065
0198	4.500	114.2	4.326		0.061		0.238		4.364		0.068		0.068
0199	4.562	116.0	4.384		0.072		0.250		4.422		0.079		0.070
0200	4.625	117.4	4.447		0.072		0.250		4.485		0.079		0.070
0201	4.687	119.0	4.508		0.072		0.250		4.547		0.079		0.070
0202	4.724	120.0	4.546		0.072		0.250		4.584		0.079		0.070
0203	4.750	120.6	4.571		0.072		0.250		4.610		0.079		0.070
0204	4.812	122.2	4.633		0.072		0.250		4.672		0.079		0.070
0205	4.875	123.8	4.695		0.072		0.250		4.735		0.079		0.070
0206	4.937	125.4	4.757		0.072	±0.004	0.250		4.797		0.079		0.070
0207	5.000	126.9	4.820		0.072		0.250		4.856		0.079		0.072
0208	5.118	130.0	4.934		0.072		0.250		4.974		0.079		0.072
0209	5.125	130.1	4.939		0.072		0.250		4.981		0.079		0.072
0210	5.250	133.2	5.064	+0.000	0.072		0.250		5.107		0.079		0.072
0211	5.375	136.5	5.187	-0.050	0.072		0.250		5.228	±0.007	0.079		0.074
0212	5.500	139.9	5.308		0.072		0.250		5.353		0.079		0.074
0213	5.511	140.0	5.320		0.072		0.250		5.364		0.079		0.074

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TABLE I. Medium duty external retaining rings - Continued.

DASH NUMBER	SHAFT DIAMETER		RING DIMENSIONS						GROOVE DIMENSIONS				
	INCHES	MM	FREE DIAMETER		THICKNESS		RADIAL WALL		DIAMETER		WIDTH		NOMINAL GROOVE DEPTH
	A		G	Tol	F	Tol	E	Tol	C	Tol	D	Tol	D_1
0214	5.625	142.8	5.433		0.072		0.250		5.478		0.079		0.074
0215	5.750	145.9	5.550		0.072		0.250		5.597		0.079		0.077
0216	5.875	149.2	5.674	+0.000	0.072		0.250	+0.004	5.722	±0.007	0.079	+0.005	0.077
0217	5.905	150.0	5.705	-0.050	0.072		0.250	-0.006	5.752		0.079	-0.000	0.077
0218	6.000	152.3	5.798		0.072		0.250		5.847		0.079		0.077
0219	6.125	155.6	5.903		0.086		0.312		5.953		0.094		0.086
0220	6.250	158.8	6.026		0.086		0.312		6.078		0.094		0.086
0221	6.299	160.0	6.076		0.086		0.312		6.127		0.094		0.086
0222	6.375	162.0	6.152		0.086		0.312		6.203		0.094		0.086
0223	6.500	165.1	6.274		0.086		0.312		6.328		0.094		0.086
0224	6.625	168.3	6.390	+0.000	0.086		0.312		6.443		0.094		0.091
0225	6.750	171.4	6.513	-0.060	0.086		0.312		6.568		0.094		0.091
0226	6.875	174.6	6.638		0.086		0.312		6.693		0.094		0.091
0227	7.000	177.8	6.761		0.086		0.312		6.818		0.094		0.091
0228	7.125	181.0	6.877		0.086		0.312	+0.004	6.933		0.094		0.096
0229	7.250	184.1	6.999		0.086		0.312	-0.008	7.058		0.094		0.096
0230	7.375	187.3	7.125		0.086	±0.004	0.312		7.183		0.094		0.096
0231	7.500	190.5	7.250		0.086		0.312		7.308		0.094		0.096
0232	7.625	193.7	7.363		0.086		0.312		7.423		0.094	+0.006	0.101
0233	7.750	196.8	7.486		0.086		0.312		7.548	±0.008	0.094	-0.000	0.101
0234	7.875	200.0	7.611		0.086		0.312		7.673		0.094		0.101
0235	8.000	203.0	7.734		0.086		0.312		7.798		0.094		0.101
0236	8.250	209.5	7.972		0.086		0.375		8.038		0.094		0.106
0237	8.500	216.0	8.220		0.086		0.375		8.288		0.094		0.106
0238	8.750	222.2	8.459	+0.000	0.086		0.375		8.528		0.094		0.111
0239	9.000	228.6	8.707	-0.070	0.086		0.375		8.778		0.094		0.111
0240	9.250	235.0	8.945		0.086		0.375		9.018		0.094		0.116
0241	9.500	241.0	9.194		0.086		0.375		9.268		0.094		0.116
0242	9.750	247.8	9.432		0.086		0.375		9.508		0.094		0.121
0243	10.000	254.0	9.680		0.086		0.375		9.758		0.094		0.121
0244	10.250	260.7	9.918		0.086		0.375		9.998		0.094		0.126
0245	10.500	265.0	10.166		0.086		0.375		10.248		0.094		0.126
0246	10.750	273.0	10.405		0.086		0.375		10.488		0.094		0.131
0247	11.000	279.4	10.653		0.086		0.375		10.738		0.094		0.131

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